Exercise Sheet 5

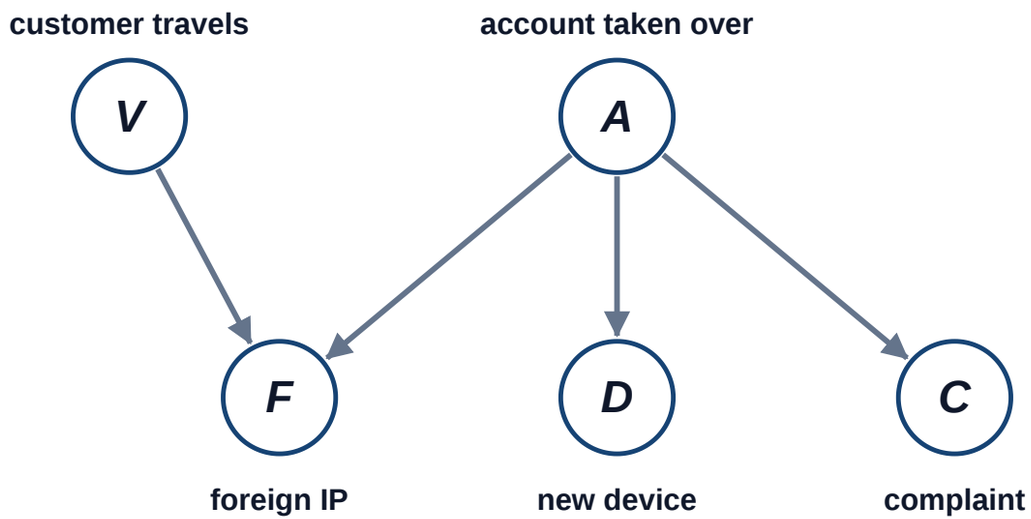
Bayesian Networks

Probabilistic Machine Learning

Prof. Dr. Michael Granitzer

5.A Graphing a Joint

Exercise 5.A.1: A Small Fraud Network



All five variables are yes/no.

1. Write the joint $p(V, A, F, D, C)$ as the network factorises it.
2. How many numbers does the network need? How many the full joint?
3. State the local Markov claim for D and for F.
4. Name one missing arrow you would challenge, and what adding it costs.

Exercise 5.A.1: A Small Fraud Network — Solution

Exercise 5.A.1: A Small Fraud Network — Solution

1. $p(V, A, F, D, C) = p(V) \cdot p(A) \cdot p(F \mid V, A) \cdot p(D \mid A) \cdot p(C \mid A)$.

2. $1 + 1 + 4 + 2 + 2 = 10$ numbers, against $2^5 - 1 = 31$.

3. D has parent A and no descendants: $D \perp\!\!\!\perp \{V, F, C\} \mid A$. F has parents V, A: $F \perp\!\!\!\perp \{D, C\} \mid V, A$.

4. For example $V \to D$: travellers often use a new phone or a hotel computer. It removes the claim $D \perp\!\!\!\perp V \mid A$ and turns $p(D \mid A)$ into $p(D \mid V, A)$: 4 numbers instead of 2, 12 in total. (Any well-argued arrow counts; the point is that each missing arrow is a claim.)

Exercise 5.A.2: Inference by Elimination

Tables: $P(V{=}1) = 0.05$, $P(A{=}1) = 0.01$; $P(F{=}1 \mid V, A)$: 0.02 for $(0,0)$, 0.9 for $(1,0)$, 0.7 for $(0,1)$, 0.95 for $(1,1)$; $P(D{=}1 \mid A)$: 0.1 for $A{=}0$, 0.6 for $A{=}1$; $P(C{=}1 \mid A)$: 0.001 for $A{=}0$, 0.5 for $A{=}1$.

1. Compute $P(F{=}1)$. Which variables can be dropped before summing, and why? Eliminate V first and give the message $m(A)$.
2. Compute $P(A{=}1 \mid F{=}1)$.
3. Compute $P(A{=}1 \mid F{=}1, D{=}1)$.
4. Compute $P(A{=}1 \mid F{=}1, V{=}1)$ and compare with 2. Name the effect.

Exercise 5.A.2: Inference by Elimination — Solution

Exercise 5.A.2: Inference by Elimination — Solution

1. D and C are unobserved leaves: $\sum_d p(d \mid A) = 1$ and $\sum_c p(c \mid A) = 1$, so they drop out. Then $P(F{=}1) = \sum_A p(A) m(A)$ with $m(A) = \sum_V p(V) P(F{=}1 \mid V, A)$: $m(0) = 0.95 \cdot 0.02 + 0.05 \cdot 0.9 = 0.064$, $m(1) = 0.95 \cdot 0.7 + 0.05 \cdot 0.95 = 0.7125$. $P(F{=}1) = 0.99 \cdot 0.064 + 0.01 \cdot 0.7125 = 0.06336 + 0.007125 = 0.070485$.

2. $P(A{=}1 \mid F{=}1) = 0.007125 / 0.070485 \approx 0.101$ — ten times the prior.

3. Multiply in $P(D{=}1 \mid A)$: $0.007125 \cdot 0.6 = 0.004275$ and $0.06336 \cdot 0.1 = 0.006336$, so $P(A{=}1 \mid F{=}1, D{=}1) = 0.004275 / 0.010611 \approx 0.403$.

4. $P(A{=}1, V{=}1, F{=}1) = 0.01 \cdot 0.05 \cdot 0.95 = 0.000475$ and $P(A{=}0, V{=}1, F{=}1) = 0.99 \cdot 0.05 \cdot 0.9 = 0.04455$, so $P(A{=}1 \mid F{=}1, V{=}1) = 0.000475 / 0.045025 \approx 0.011$ — back to about the prior. Travel **explains away** the foreign IP: V and A are independent, but not given their common effect F.

5.B Reading Independence

Exercise 5.B.1: d-Separation

For the fraud network of 5.A.1, decide with d-separation — give the path and the triple that blocks or opens it:

1. $V \perp\!\!\!\perp A$?
2. $V \perp\!\!\!\perp A \mid F$?
3. $V \perp\!\!\!\perp D$?
4. $V \perp\!\!\!\perp D \mid F$?
5. $V \perp\!\!\!\perp D \mid F, A$?
6. $D \perp\!\!\!\perp C$? and $D \perp\!\!\!\perp C \mid A$?

Then check 3 and 4 with the numbers of 5.A.2: compare $P(V=1 \mid F=1)$ with $P(V=1 \mid F=1, D=1)$.

Exercise 5.B.1: d-Separation — Solution

Exercise 5.B.1: d-Separation — Solution

query	path	verdict
$V \perp\!\!\!\perp A$	$V \to F \leftarrow A$: collider, unobserved	**yes**
$V \perp\!\!\!\perp A \mid F$	the collider is observed: open	**no**
$V \perp\!\!\!\perp D$	$V \to F \leftarrow A \to D$: collider F unobserved	**yes**
$V \perp\!\!\!\perp D \mid F$	collider F open, fork A unobserved: open	**no**
$V \perp\!\!\!\perp D \mid F, A$	fork A observed: blocked	**yes**
$D \perp\!\!\!\perp C$	$D \leftarrow A \to C$: fork, unobserved	**no**
$D \perp\!\!\!\perp C \mid A$	fork observed	**yes**

Numbers. $P(V{=}1) = 0.05 = P(V{=}1 \mid D{=}1)$ (claim 3). Given $F{=}1$: $P(V{=}1 \mid F{=}1) = 0.045025/0.070485 \approx 0.639$, but $P(V{=}1 \mid F{=}1, D{=}1) = 0.00474/0.010611 \approx 0.447$: a new device makes a take-over more likely, which explains the foreign IP and makes travel less likely. Claim 4 fails, as d-separation said.

Exercise 5.B.2: Which Arrows Are in the Data?

1. Which of these graphs are Markov equivalent? (a) $X \to Y \to Z$ · (b) $X \leftarrow Y \leftarrow Z$ · (c) $X \leftarrow Y \to Z$ · (d) $X \to Y \leftarrow Z$ · (e) (a) plus $X \to Z$
2. In the fraud network of 5.A.1, which single arrows can be reversed without changing any independence claim? Can $A \to D$ and $A \to C$ be reversed together?
3. An analyst fits the fraud network from data and says "take-overs **cause** new devices". What can the data support?

Exercise 5.B.2: Which Arrows Are in the Data? — Solution

Exercise 5.B.2: Which Arrows Are in the Data? — Solution

1. (a), (b), (c) share the skeleton $X - Y - Z$ and have no collider: all claim $X \perp\!\!\!\perp Z \mid Y$ and nothing else — **equivalent**. (d) has the collider at Y: it claims $X \perp\!\!\!\perp Z$ instead — different. (e) is complete and claims no independence at all — different from all four.

2. Equivalent graphs need the same skeleton and the same colliders. The collider $V \to F \leftarrow A$ must stay, so $V \to F$ and $A \to F$ cannot be reversed. $A \to D$ alone can be reversed ($D \to A \to C$ creates no collider), and so can $A \to C$ alone. Both together give $D \to A \leftarrow C$, a new collider — not equivalent.

3. Only that A and D are dependent, and — since $A \to D$ is reversible — not its direction. The causal reading comes from knowledge outside the data (a take-over happens before the device is used), not from the fit.

5.C Plates and Learning

Exercise 5.C.1: Naive Bayes as a Plate Diagram

Naive Bayes for N transactions with p yes/no flags: class prior $\pi = P(y = 1)$ with a $\mathrm{Beta}(a, b)$ prior; flag probabilities $\theta_{jc} = P(x_j = 1 \mid y = c)$, each with a $\mathrm{Beta}(1, 1)$ prior.

1. Draw the plate diagram: nodes, shading, dots, plates (and which plates are nested).
2. Write the factorisation that the diagram encodes.
3. At prediction time a new transaction arrives with flags but no label. What changes in the drawing?

Exercise 5.C.1: Naive Bayes as a Plate Diagram — Solution

Exercise 5.C.1: Naive Bayes as a Plate Diagram — Solution

1. Dot (a, b) → open node π → shaded y_i. A plate over N transactions holds y_i and, **nested** inside it, a plate over the p flags holding the shaded x_{ij}. Outside the N plate: open nodes θ_{jc} in a plate over the p flags and a plate over the 2 classes, fed by a dot $(1, 1)$. Arrows $y_i \to x_{ij}$ and $\theta_{jc} \to x_{ij}$ (y_i selects which $\theta_{j\cdot}$ applies).

2.

$$p(\pi \mid a, b) \prod_{j=1}^{p} \prod_{c=0}^{1} p(\theta_{jc}) \prod_{i=1}^{N} \left[p(y_i \mid \pi) \prod_{j=1}^{p} p(x_{ij} \mid y_i, \theta_{j0}, \theta_{j1}) \right]$$

3. The new label y_{new} is **unshaded** — it is inferred. The parameters are either conditioned on the training data (the full Bayesian answer) or replaced by their smoothed counts — the posterior means of Lecture 03.

Exercise 5.C.2: Counting, Elimination, and an Empty Row

1. A chain $X_1 \to \cdots \to X_{10}$ of yes/no variables. How many terms does $p(x_{10})$ need if all other variables are summed at once, and roughly how many operations by elimination?
2. From 200 records of the fraud network (table), estimate $P(F{=}1 \mid V, A)$ by maximum likelihood and with one pseudo-count per value. What happens to the last row, and what does smoothing put there?

V	A	records	with $F = 1$
0	0	180	4
1	0	15	13
0	1	5	3
1	1	0	0

Exercise 5.C.2: Counting, Elimination, and an Empty Row — Solution

Exercise 5.C.2: Counting, Elimination, and an Empty Row — Solution

1. $2^9 = 512$ terms (each a product of 10 factors), against about $4 \cdot 9 = 36$ multiplications by passing a two-entry message along the chain.

2.

V, A	MLE	smoothed $(n_1 + 1)/(n + 2)$
$0, 0$	$4/180 \approx 0.022$	$5/182 \approx 0.027$
$1, 0$	$13/15 \approx 0.867$	$14/17 \approx 0.824$
$0, 1$	$3/5 = 0.6$	$4/7 \approx 0.571$
$1, 1$	$0/0$ — undefined	$1/2 = 0.5$

A parent configuration that never occurred leaves its row **undetermined** under maximum likelihood; the pseudo-counts fill it with the prior mean $1/2$. Sparse rows are one more reason to keep the number of parents small.